THE ESTIMATOR DECIDES: FINITE-GROUP ACTIVITY, CURRICULUM ALLOCATION, AND HINDSIGHT COVERAGE IN RL WITH VERIFIABLE REWARDS

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ABSTRACT

Difficulty curricula and hindsight relabeling are often treated as objective-agnostic modules for reinforcement learning with verifiable binary rewards. We show that their effect depends on the finite-group estimator underneath. For the practical centered MaxRL implementation that drops all-fail groups, a prompt with pass rate p and N rollouts has expected coefficient mass $A_N(p) = 2[(1-p) - (1-p)^N]$. Its expected update factorizes as $\mathbb{E}[\widehat{g} \mid x] = \frac{1}{2} A_N(p)(\mu_+ - \mu_-)$, so the normalized mass is exactly the estimator-controlled scalar multiplying the success-versus-failure score separation. Its shape and unique peak are set by N, motivating a prompt sampler without a hand-set target difficulty. The result is implementation-specific: raw MaxRL and the full control-variate variant target order N, whereas the deployed drop-all-fail estimator targets order $N-1$; the full variant assigns negative coefficients when every rollout fails. In a matched-budget 20-seed tabular control, that alternative invokes the update callback for every all-fail group but does not reliably bootstrap the frontier (its median rounds to zero), while verified hindsight reaches 0.927 ± 0.003 AUC. Relabeling, however, produces a verified destination-conditioned update under a source-induced proposal, not generally an on-policy destination gradient. In historical Countdown runs, three-seed aggregate endpoints for a legacy per-trace relabeler show higher mean success and lower pass@16 coverage; its saturation gate has one promising but unintended moderate operating point. Historical maze runs use a legacy utility and isolated positive imitation updates. They show an estimator-associated coverage separation, but neither neural suite instantiates the proposed common-destination algorithm, and the maze's unbalanced sampler mix and shared warm starts preclude a causal interaction claim. Together the theory and experiments delimit what allocation can change, what relabeling can create, and why both mean success and pass@k coverage must be reported.

1 INTRODUCTION

Post-training with binary verifiers is constrained by a simple event: a centered group update needs reward contrast. If all N rollouts fail or all N succeed, the practical drop-constant-group estimator is silent. This finite-group fact is not visible in a population weight function, which describes a prompt only after averaging over sampling. A curriculum acts one step earlier, deciding which prompts receive the rollout budget needed to make those weights observable.

The field's response is a pair of data-level interventions. The first is the *difficulty curriculum*: reallocate the rollout budget toward tasks where learning is possible right now — UCB bandits over difficulty buckets (Wang et al., 2025), target-difficulty controllers (Shi et al., 2026), category bandits (Chen et al., 2025), learnability-based rejection sampling (Foster et al., 2025), dead-prompt filtering (Yu et al., 2025; Zheng et al., 2025). The second is *failure recycling*: relabel a failed rollout as a verified example of an achieved destination (Andrychowicz et al., 2017; Ghosh et al., 2021; Xu et al., 2026; Zhang et al., 2026). Both interventions are often evaluated as estimator-agnostic modules, and mainly with mean accuracy. That can miss two distinct questions: whether the deployed estimator produces a usable finite-group update, and whether concentrating updates changes multi-sample coverage.

We study these questions through the exact finite-group coefficient profile of the estimator actually deployed. For practical centered MaxRL, its normalized coefficient mass has two endpoint zeros and one interior maximum. These induce three *operational*, budget-dependent regimes: negligible contrast, a frontier of high estimator activity, and a saturated tail. The profile constrains local activity; it does not by itself determine cross-task transfer or final performance. Three questions organize the paper:

Q1: What activity profile does the estimator implement? For the practical centered/drop estimator, the expected coefficient mass is $A_N(p) = 2(\text{pass@}N - \text{pass@}1)$ and the expected update factorizes into $A_N/2$ times the success-versus-failure score separation (§2). The curve's shape and exact peak follow from N; under our normalization, its $N{=}2$ case is the $p(1-p)$ learnability score used by prior curricula. Raw MaxRL and full-control-variate MaxRL have different activity profiles, so the result belongs to the deployed implementation, not to every estimator bearing the MaxRL name.

Q2: What can allocation and relabeling change? Reallocation cannot manufacture a positive anchor for an all-fail group under the practical estimator. The full control-variate estimator is an important counterexample at the sample level: it assigns a negative-only coefficient vector and invokes the update path there. We therefore test it directly rather than claim impossibility. Hindsight can instead create verified destination contrast, but its data are drawn from a source-induced proposal and are generally off-policy for the destination (§4). Historical Countdown runs used a different, per-trace relabeler; their three-seed aggregate endpoints show higher mean accuracy and lower pass@k coverage, a descriptive pattern we call **recycling-induced sharpening** rather than validation of the proposed destination-group estimator.

Q3: Can destination saturation control that trade? The mastered endpoint motivates rejecting relabels to already-reachable destinations. We instantiate this with a tuned pass-rate threshold. At an unintended moderate effective strength, the three-seed frontier-coverage point estimate returns near baseline while retaining part of the mean-gain point estimate; the corrected stronger setting is only a single-seed endpoint and is reported as preliminary (§6.9). Thus the algebra motivates the admission variable, while the operating threshold remains empirical.

Our contributions:

1. **Finite-group theory.** We distinguish raw, full-control- variate, and practical MaxRL; derive the practical estimator's exact coefficient-mass profile and update factorization; and show that its drop-all-fail convention corresponds to truncation order $N-1$.

2. **Allocation and recycling.** We use normalized coefficient mass as a prompt-sampling utility and characterize hindsight as a verified but generally off-policy destination semi-gradient.

3. **Missing-control evidence.** A matched-budget 20-seed CPU experiment shows that the full-control-variate all-fail coefficient branch does not reliably unlock the frontier-heavy pool, whereas exact relabeling does. A separate fixed-schedule control with disjoint tuning and held-out evaluation seeds puts practical MaxRL, GRPO, and RLOO within 0.005 AUC on the balanced exact-score pool.

4. **Coverage effects and boundaries.** Legacy Countdown runs report a three-seed aggregate mean-versus-pass@k cost for their per-trace relabeler; a saturation gate is promising but preliminary. Maze and LLM runs are retained as explicitly implementation-mismatched diagnostics, while locomotion and Jugs mark further evidence limits.

The order-$N-1$ result refines the implementation-level reading of MaxRL Algorithm 1; it does not contradict the order-N theorem for the raw or full-control-variate estimators. Negative results and retractions are reported in place (§7).

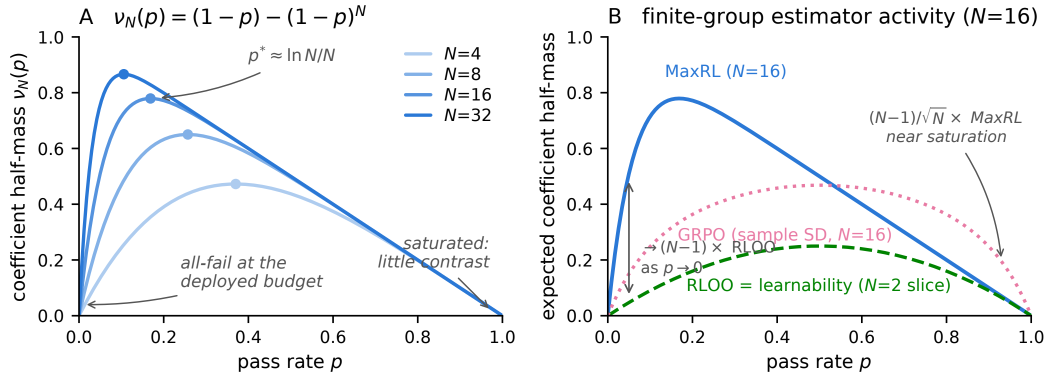

Figure 1: **Finite-group estimator activity.** *Left:* normalized coefficient mass $\nu_N(p) = (1-p) - (1-p)^N$ for the practical centered/drop estimator. Its exact peak is $p^* = 1 - N^{-1/(N-1)}$; low activity near an endpoint is operationally important only relative to the deployed budget. *Right:* expected coefficient half-mass at N=16. The GRPO curve uses the code's sample-standard-deviation convention (fixed 10^{-6} stabilizer included), not the population-SD limit. The following endpoint ratios take the zero-stabilizer limit, as in Proposition 2; the stabilizer changes no plotted digit. As $p \to 0$, practical MaxRL/RLOO tends to $N-1$ and practical MaxRL/GRPO tends to $\sqrt{N}$; as $p \to 1$, MaxRL and RLOO both decay linearly while GRPO/MaxRL tends to $(N-1)/\sqrt{N}$. Under our normalization, RLOO is proportional to the $p(1-p)$ learnability score used by SFL and LILO.

Table 1: **Three MaxRL estimators.** Mass is expected total absolute score coefficient, $A(p) = \mathbb{E}\sum_i |w_i|$. The order-$N-1$ statement applies only to the practical convention that drops the whole all-fail group.

estimator	$K=0$ sample update	target	coefficient mass $A(p)$
raw H_N	0	J_N	$1-q^N$
full CV C_N	$-N^{-1}\sum_i S_i$	J_N	$2q-q^N$
practical D_N	0	J_{N-1}	$2(q-q^N)$

2 From MaxRL objectives to finite-group activity

For a prompt x and group size $N \geq 2$, let $r_i \in \{0,1\}$, $K = \sum_{i=1}^N r_i$, $p = \Pr(r=1 \mid x)$, $q = 1-p$, and $S_i = \nabla_\theta \log \pi_\theta(z_i \mid x)$. MaxRL truncates $\log p = -\sum_{t \geq 1} q^t/t$ at

$$J_T(p) := -\sum_{t=1}^{T} \frac{q^t}{t}.$$

The baseline paper separates a raw estimator from its control-variate form; its practical algorithm adds a further constant-group convention. Those three choices have different finite-group activity profiles:

$$H_N := \mathbf{1}\{K>0\}\frac{1}{K}\sum_i r_i S_i, \qquad \text{raw success average,} \tag{1}$$

$$C_N := H_N - \frac{1}{N}\sum_i S_i, \qquad \text{full control variate,} \tag{2}$$

$$D_N := \mathbf{1}\{K>0\}\left(\frac{1}{K}\sum_i r_i S_i - \frac{1}{N}\sum_i S_i\right), \qquad \text{practical centered/drop.} \tag{3}$$

Tajwar et al.'s order-N theorem is exact for H_N, and the zero-mean control variate leaves C_N unbiased for the same target (Tajwar et al., 2026). The next lemma concerns only the additional drop rule in the shipped practical loop.

Lemma 1 (Order of the practical drop estimator)**.** *The practical estimator D_N is unbiased for J_{N-1}, not J_N.*

Proof. Relative to C_N, zeroing the control variate only on $K = 0$ removes an expected $q^{N-1}\nabla p$. Hence

$$\mathbb{E}[D_N \mid x] = \left(\sum_{j=0}^{N-1} q^j - q^{N-1}\right)\nabla p = \sum_{j=0}^{N-2} q^j \nabla p = \nabla J_{N-1}(p).$$

□

This is an implementation-level refinement, not a correction of the raw estimator theorem.

Proposition 1 (Coefficient mass and exact update factorization)**.** *For $0 < p < 1$, define*

$$\mu_+(x) = \mathbb{E}[S \mid r = 1, x] = \frac{\nabla p}{p}, \qquad \mu_-(x) = \mathbb{E}[S \mid r = 0, x] = -\frac{\nabla p}{q}.$$

For the practical estimator,

$$\boxed{A_N(p) := \mathbb{E}\sum_i |w_i| = 2\nu_N(p), \qquad \nu_N(p) := q - q^N = \textit{pass@}N(p) - p,} \tag{4}$$

and

$$\boxed{\mathbb{E}[D_N \mid x] = \nu_N(p)(\mu_+ - \mu_-) = \frac{\nu_N(p)}{pq}\nabla p = \sum_{j=0}^{N-2} q^j \nabla p.} \tag{5}$$

Proof. Conditioned on $K = k > 0$, the positive coefficients sum to $1 - k/N$, and the negative coefficients have the same absolute sum. Thus the conditional coefficient mass is $2(1 - k/N)$ and the conditional expected update is $(1 - k/N)(\mu_+ - \mu_-)$. Averaging over $K \sim \mathrm{Bin}(N, p)$ gives $\mathbb{E}[(1 - K/N)\mathbf{1}\{K > 0\}] = 1 - q^N - p = q - q^N$. Finally, $\mu_+ - \mu_- = \nabla p/(pq)$. □

Equation 5 gives ν_N a precise interpretation: it is the estimator-controlled scalar multiplying the success-versus-failure score separation. It is not the entire gradient geometry. Its unique maximum is

$$p^\star = 1 - N^{-1/(N-1)} \approx \frac{\log N}{N},$$

and its endpoint zeros induce operational regimes rather than mathematically fixed-width bands. If a task receives B groups, its expected number of nonconstant groups is

$$B\,[1 - p^N - q^N].$$

We call it operationally inactive at budget (B, N) when this quantity is much smaller than one. At $p > 0$, increasing B or N can move the task out of that regime; at $p = 0$, reallocation under the same policy and verifier cannot produce a success.

Proposition 2 (Finite-group comparisons)**.** *Under the coefficient conventions used here, RLOO has normalized half-mass $\nu^{\mathrm{RLOO}}(p) = pq$, so $\nu_N/\nu^{\mathrm{RLOO}} \to N - 1$ as $p \to 0$, while both decay as q when $p \to 1$. For the deployed sample-standard-deviation GRPO implementation,*

$$\nu_N^{\mathrm{GRPO}}(p) = \mathbb{E}\left[\frac{K(N-K)}{N^2\{\sqrt{K(N-K)/(N(N-1))} + \epsilon\}}\right], \tag{6}$$

with constant groups contributing zero. At $\epsilon = 0$, this is $\sqrt{(N-1)/N}\,N^{-1}\mathbb{E}\sqrt{K(N-K)}$, giving

$$\frac{\nu_N}{\nu_N^{\mathrm{GRPO}}} \to \sqrt{N} \quad (p \to 0), \qquad \frac{\nu_N^{\mathrm{GRPO}}}{\nu_N} \to \frac{N-1}{\sqrt{N}} \quad (p \to 1).$$

The population-SD GRPO curve often used in objective analyses differs by the finite-N factor above; Eq. 6 matches our code and historical experiments. The 10^{-6} stabilizer changes no reported digit at the group sizes studied.

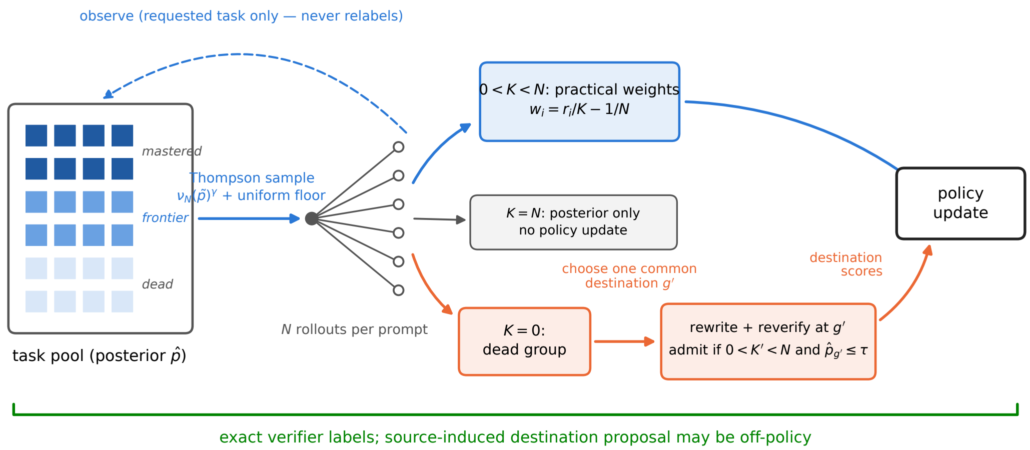

Figure 2: **Proposed FrontierMax step.** The teacher samples requested prompts by ν_N. A live requested group uses the practical estimator. For an all-fail group, the relabeler selects one common destination, rewrites and reverifies the group, and updates only if the destination group has contrast $0 < K' < N$. The resulting update is verified but generally off-policy for the destination. Posterior state is updated only from requested-task outcomes.

Proposition 3 (One-step fixed-rate allocation)**.** *Suppose current task pass rates p_i are known and held fixed, task i receives an integer group size $n_i \geq 1$, and $\sum_i n_i = B$. Greedy water-filling maximizes the separable one-step objective $\sum_i \nu_{n_i}(p_i)$: the marginal value of the next rollout is*

$$\nu_{n_i+1}(p_i) - \nu_{n_i}(p_i) = p_i(1-p_i)^{n_i},$$

which decreases with n_i.

This proposition is a mass-allocation oracle under fixed p_i, not a ceiling on sequential AUC or held-out performance. Policy change, cross-task transfer, interference, posterior uncertainty, exploration floors, and delayed skill unlocking all lie outside its assumptions. The deployed teacher keeps a fixed group size and changes which prompt group is drawn, using ν_N^γ with a uniform floor.

Remark 1 (Activity is not transfer). Equation 5 is exact for a prompt-local expected update, but neither coefficient mass nor its endpoint zeros determine gradient norm, cross-task transfer, or final performance. On the exact-score bridge, ν_N closely ranks first-order improvement (`curriculum_maxrl/results_bridge.json`); later planning experiments show that the best sequential index depends on whether the target is anytime AUC or a final checkpoint. We therefore use ν_N as an estimator-aligned sampling score and test downstream performance rather than treating it as a universal learning utility.

Remark 2 (Adaptive sampling changes the training distribution). Conditioned on training history and treating selection as stop-gradient, sampling prompts from q_t yields $\mathbb{E}_{x\sim q_t}[\nabla J(x)]$, not the data-distribution gradient under ρ unless corrected. Because q_t adapts, these updates are not generally the gradient of one fixed reweighted objective. Evaluation definitions are fixed and method-independent under ρ. The exact-score controls reuse their task pool; disjoint tuning and evaluation seeds are used where stated. Gym uses fixed task bins with fresh episodes rather than a held-out task pool.

The theory separates two interventions. Allocation changes q_t to spend groups where the practical estimator is active. Hindsight changes the conditioning and the destination proposal, potentially creating verified contrast where the requested group has none. The latter falls outside Proposition 1 and requires its own distributional analysis.

3 FrontierMax: Estimator-Aligned Allocation

FrontierMax is the corrected, proposed group-rollout loop whose teacher tracks requested-task pass rates with decayed Beta state. Thompson sampling draws $\tilde{p}_x$ and samples prompt groups proportional

Algorithm 1 FrontierMax: one group step.

1: ▷ sample requested prompts $x_1, \ldots, x_B$ proportional to $\nu_N(\tilde{p}_x)^\gamma$, with a uniform floor; $\tilde{p}_x \sim \mathrm{Beta}(\alpha_x, \beta_x)$
2: roll out N completions per prompt; verify requested rewards r_i; set $K = \sum_i r_i$
3: update (α_x, β_x) from requested outcomes only
4: if $0 < K < N$, update with $w_i = r_i/K - 1/N$
5: ▷ if $K = 0$, choose one destination $g' = H(x, z_{1:N})$; rewrite every conditioning to g'
6: reverify all rewritten trajectories under g', obtaining r'_i and $K' = \sum_i r'_i$
7: ▷ admit only if $0 < K' < N$ and $\hat{p}_{g'} \le$ `gate_max_p`; recompute destination scores and use $w'_i = r'_i/K' - 1/N$
8: update the policy from live requested groups and admitted destination groups

to $\nu_N(\tilde{p}_x)^\gamma$, mixed with a uniform floor. Relabeled outcomes never update the requested-task posterior; doing so inflated $\hat{p}$ from .47 on evaluation to .81 in a legacy maze control. The profile ν_N, its peak, and its N-dependence are derived. Posterior decay, the floor, concentration γ, relabel dose, and gate threshold are tuned controls (Table 4), not consequences of the theorem.

A per-prompt posterior is appropriate only when prompts recur enough to accumulate evidence. At pools much larger than the number of groups, our legacy-utility GSM8K pilot remains nearly uniform; a bucket posterior or learned difficulty model is required at that scale. The current method claim is therefore about repeated finite pools, with the large-pool result treated as a delivery diagnostic.

Algorithm 1's two defining pieces—the exact ν_N sampler and a common-destination, mixed-outcome relabeled group—are instantiated together in the exact-score chain and in the corrected NumPy interfaces. GridReach and Countdown regression-test the shared-destination contrast explicitly; chain and Gym controls exercise relabeling without asserting every contract in a dedicated regression. The historical neural and LLM runs predate this contract: maze uses the legacy $(1 - q^N)q$ utility and per-trajectory positive-only recycling; GSM8K uses the legacy utility; and Countdown combines that utility with per-trace targets inside a uid group. Those runs are retained below as diagnostics, not as end-to-end validation of Algorithm 1.

The gate is saturation-motivated, not parameter-free. With the $\mathrm{Beta}(1, 1)$ initialization and threshold 0.5, an unseen destination begins exactly on the admission boundary; the hard rule does not model cold-start uncertainty. More conservative posterior bounds or held-out destination probes are natural extensions.

4 Failure recycling and proposal shift

Let $h_{\theta,g}(Z_{1:N})$ denote the practical destination update after conditioning has been rewritten to g and every trajectory has been reverified. Fresh destination groups follow

$$P_{\theta,g} = \pi_\theta(\cdot \mid g)^{\otimes N}, \qquad \nabla J_{N-1,g} = \mathbb{E}_{P_{\theta,g}}[h_{\theta,g}].$$

Hindsight instead samples a source prompt x, chooses a common destination from the group, rewrites the group, and conditions on admission. Let $Q_{\theta,g}^{x \to g}$ be that induced proposal law. Its expected update is

$$U_g^{\mathrm{HS}} = \mathbb{E}_{Q_{\theta,g}^{x \to g}}[h_{\theta,g}].$$

Exact verification makes the label semantically valid; it does not make $Q_{\theta,g}^{x \to g} = P_{\theta,g}$. Equality of the two update moments is necessary and sufficient for an exact fresh-destination gradient, and equality of the laws is sufficient. Otherwise hindsight is a distribution-shifted destination semi-gradient because the optimization does not differentiate through the source proposal, destination choice, or admission event. Same-group destination selection also breaks the fixed-destination iid premise; selecting g with one group and updating from a disjoint group is the clean cross-fitted control. This framing matches the off-policy gap in GCSL and the corrected relabel distribution derived by hindsight inference for policy improvement (Ghosh et al., 2021; Eysenbach et al., 2020).

Three implementation contracts remain essential: **(1) semantic verification** under the destination's exact verifier; **(2) one common destination**, with every conditioning and score recomputed under that

Table 2: Claims and evidence strength. "Blocks" are independent training seed/warm-start units; heterogeneous historical runs are not counted as independent factorial replicates.

rung	setting	independent units	supported claim
6.1–6.2	exact-score skill chains	20 primary; 10 tune + 20 held-out	full CV unreliable; tuned fixed-schedule estimators within .005; hindsight unlocks
6.3–6.4	1.26M maze transformer	3 shared blocks; heterogeneous arms	legacy utility/recycler; descriptive estimator-associated coverage pattern
6.5	IsaacLab Anymal-C	1 seed	hypothesis-consistent pilot only
6.6	Gym binary-success control	3 paired seeds	shared transfer and task spread matter
6.7	GSM8K 360M pilot	1–2 seeds/cell	legacy-utility treatment-delivery diagnosis
6.8–6.9	Countdown v2	3 paired seeds	legacy per-trace sharpening; preliminary gate
limits	Jugs feasibility screen	one committed screen	boundary hypothesis only

destination; and **(3) destination contrast**, $0 < K' < N$. The gridworld adapter and regression tests enforce rewriting and reverification. An older no-rewrite comparison is described in project notes, but its result artifact is not vendored and is therefore not used quantitatively here. These behavioral contracts do not establish on-policy exactness.

The historical fixed-pool placebo battery suggests that much of hindsight's $+0.22$ AUC gain is compatible with receiving an additional update: live-gradient replay reaches 0.832, versus 0.869 for relabeling and 0.650 for uniform training. The $+0.037$ hindsight-minus-replay difference is consistent across five seeds. However, the runs did not log per-arm update counts or update norms, and the random-target arm reused a stale reward vector with an unseeded random draw rather than exactly re-verifying the destination. The residual is therefore a behavioral comparison, not a causal direction estimate or a clean dose decomposition. The checkout also contains no exact-score streaming arm, so whether the reusable finite pool is necessary for the gain remains open.

5 Experiments: an escalating ladder

Table 2 separates evidence by experimental unit and claim strength. Unless stated otherwise, $\pm$ is sample standard deviation across independent training seeds; prompts, levels, checkpoints, values of k, and repeated evaluations are repeated measures, not extra seeds. Paired contrasts share a seed and warm start. Coverage is pass@k at the suite-specific k (maze 8, Countdown 16, GSM8K 4). Exact-score and LLM fixed-grid AUC is the unweighted mean over the reported evaluation checkpoints. Historical maze AUC is normalized trapezoidal optimizer-step AUC on each run's own logged grid; matched wall-clock is the stopping budget, not the AUC abscissa. We distinguish training-seed variation from fixed-checkpoint evaluation noise. The historical maze labels are not exchangeable, so earlier pooled permutation p-values are withdrawn; that evidence is descriptive.

6.1 Exact-score chains: what comes from allocation, dose, and direction? On the tabular shared-skill chain, uniform sampling reaches 0.650 AUC and the posterior ν_N teacher reaches 0.728. A five-seed historical control battery reaches 0.798–0.800 under double learning rate or the random-target placebo, 0.832 under live-gradient replay, and 0.869 under verified relabeling. The hindsight-minus-replay difference is $+0.037$ in all five paired seeds. Because update counts/norms were not logged and the random-target arm was not exactly re-verified, these arms suggest—but do not causally decompose—dose and direction. The order-agnostic $p(1-p)$ teacher forfeits the allocation gain ($-.007$ versus uniform), whereas ν_N adds $+.078$, five of five paired.

A true-pass-rate, no-floor, γ=4 utility benchmark reaches 0.8885; the full posterior-plus-recycling stack reaches 0.8895, and true-pass-rate allocation plus recycling reaches 0.8935. These nearby numbers compare different interventions and do *not* establish a sequential allocation ceiling. In a matched γ=1 sampler comparison, the true-rate oracle reaches 0.8511 versus 0.7278 for the posterior teacher, showing the value of exact rates on this chain under that protocol. Relabeling also retains an

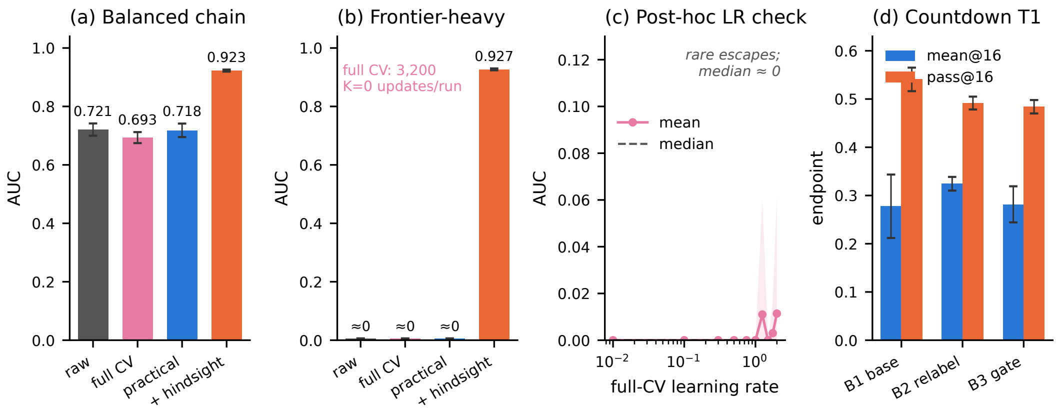

Figure 3: **Local estimator controls and sensitivity.** Panels (a–b) compare the three MaxRL variants and practical+hindsight at a common 3,200-group budget over 20 paired seeds. Full CV assigns negative coefficients and invokes the update callback on every frontier all-fail group but does not reliably bootstrap; hindsight does. Panel (c) is a post-hoc full-CV learning-rate sensitivity: rare large-step escapes do not form a reliable regime. Panel (d) shows the three-seed legacy per-trace Countdown mean/coverage trade; B3 is the unintended moderate gate operating point. All values load from committed JSON artifacts.

incremental effect at the true-rate operating point. All arms finish within .027, so the interventions mainly buy time-to-ceiling in this reusable fixed pool.

Takeaway. Coefficient-mass allocation helps on the chain. A historical placebo battery is consistent with a substantial extra-update contribution, and hindsight remains $+.037$ above live-gradient replay, but incomplete update logging prevents a causal dose–direction decomposition.

6.2 Can a negative-only all-fail update bootstrap the frontier? (20 paired seeds; exact-score tabular testbed). This is the missing estimator control suggested by the MaxRL taxonomy. We compare raw H_N, full-CV C_N, practical D_N, and practical plus exact prefix relabeling under identical uniform task draws, rollout random numbers, common learning rate 0.5, and 3,200 groups of $N = 16$. The frontier-heavy pool contains levels 5–12 and has maximum initial pass rate 10^{-5}. Full CV assigns nonzero negative coefficients and invokes `policy.update` on every one of the 3,200 all-fail groups per run, yet does not reliably bootstrap: its AUC is 1.74×10^{-6} (reported as 0.000), comparable to raw and practical. Verified hindsight uses 158.0 relabeled groups on average and reaches 0.9269 ± 0.0032 AUC / 0.9788 ± 0.0006 final, winning all 20 paired seeds. On the balanced pool, full CV reaches 0.6934 ± 0.0189, practical reaches 0.7183 ± 0.0231, and practical is higher by .0249 in 16/20 paired seeds; hindsight reaches 0.9227 ± 0.0035.

This is a budget- and testbed-specific negative result for the alternative, not an impossibility theorem. A post-hoc full-CV learning-rate sweep finds one of 20 seeds exceeding final pool mean .1 at each of rates 1.25 and 2.0, none at the other eight rates, and median AUC approximately zero throughout (best mean AUC .011). These rare, non-monotone escapes remain far below hindsight and do not define a reliable operating region. The primary comparison used no estimator-specific tuning. In an earlier five-seed regime control, uniform, DAPO-style redraw, and posterior allocation also round to zero, whereas uniform+recycling and teacher+recycling have similar observed AUCs (.931 and .928). Thus the evidence supports a narrow statement: in this frontier, redistributing negative score mass did not replace a verified positive anchor at the matched budget.

The common-rate comparison is not a fair performance ranking of differently scaled estimators. We therefore replay an identical uniform task sequence, group size, and rollout-uniform stream for practical MaxRL, GRPO, and RLOO. Learning rates are selected by a one-standard-error rule on 10 tuning seeds and evaluated on 20 separate held-out seeds. Practical MaxRL reaches 0.9889 ± 0.0089 evaluation-grid AUC, GRPO 0.9841 ± 0.0121, and RLOO 0.9869 ± 0.0099; the largest mean separation is .0048. On binary mixed groups, these estimators' coefficient vectors are positive scalar multiples of the same centered reward vector, with a K-dependent scale. This local near-ceiling control shows why common-rate comparisons can exaggerate scale differences. Its ordering is specific to the parameterization-dependent one-standard-error rule, so we do not treat it as an intrinsic estimator ranking or evidence about the adaptive neural-maze interaction.

Takeaway. Full CV proves that nonzero coefficients can reach the policy update callback on all-fail samples; the 20-seed control shows that this branch does not reliably unlock this frontier, while verified relabeling does. Fixed-schedule tuning also puts three collinear binary-reward estimators within .005 AUC here.

6.3 Historical neural maze: legacy utility/recycling and a descriptive estimator-associated coverage pattern (1.26M parameters). The matched-wall-clock suite contains roughly 30 historical runs across sampler, estimator, relabel dose, and utility-form ablations. Its wall-clock limit defines each run's stopping budget, while the reported AUC integrates over that run's optimizer-step grid. The primary method comparison is three paired warm-start blocks: the legacy frontier-ALP teacher plus dense per-trajectory positive-only recycling reaches 0.252 ± 0.006 final / 0.229 ± 0.010 AUC, versus uniform practical MaxRL at 0.230 ± 0.018 / 0.211 ± 0.014; each block's final and AUC difference is positive. In the separate teacher-only pairs, frontier-ALP arms log 1.2–$1.4\times$ more optimizer steps per wall-clock budget. The cause is unresolved: generated actions/tokens and runtime composition were not logged, and zero-weight-group counts do not support attributing the difference to skipped backward passes. At a common step count, teacher-only adds .006 AUC with mixed paired signs, whereas teacher+recycling remains positive in all three blocks. The equal-step result attenuates the clock-matched effect and flips one paired sign, suggesting that runtime composition contributes; with three blocks and no generated-token counts, its share is unidentified. The positive-only recycler is the more stable residual pattern in this archive. These runs use neither the exact ν_N sampler nor a common-destination contrastive group and therefore do not validate Algorithm 1.

For the estimator comparison, the frozen selected endpoint cohort contains five practical-MaxRL-labelled runs (two uniform, three frontier-ALP) and four GRPO-labelled runs (three uniform, one frontier). Their Δpass@8 values are all positive for the MaxRL-labelled runs and all negative for the GRPO-labelled runs. This is an informative historical pattern, but sampler composition is unbalanced and seed/warm-start blocks recur. Neither the earlier unrestricted calculation nor its sampler-stratified replacement has a valid exchangeable reference distribution; both are withdrawn as inferential claims. The suite supports an *estimator-associated* coverage separation across selected curriculum variants, not a causal teacher-by-estimator interaction. A balanced factorial with frozen realized schedules remains required.

The CPU utility-form artifact compares exact ν_N, the legacy frontier form, and a $p(1-p)$ slice. The corresponding three-seed maze verdict file and raw utility-form logs are not vendored, so the prior maze ranking is withdrawn. We therefore claim the activity profile and peak, not a universally optimal within-frontier ranking. The single-checkpoint inference sweep is retained as a secondary diagnostic in Appendix A: GRPO leads at pass@1, the summed-coverage curves cross near $k = 4$, and practical MaxRL leads at larger k.

6.4 Exploratory localization across the maze archive. Figure 4 uses a frozen list of four GRPO and 18 nonempty practical-MaxRL logs. These are heterogeneous ablations, not 22 independent replicates of one treatment. The exact cohort and explicitly unrecorded metadata are committed in `fig8_maze_run_registry.csv`. Descriptively, GRPO-labelled pass@8 loss concentrates on levels 1–3 while pass@1 there rises; practical-MaxRL logs are approximately flat on levels 1–3 and gain more on levels 4–6. The GRPO coverage gap $\overline{\text{pass@8} - \text{pass@1}}$ erodes early: more than half of its endpoint change appears by one quarter of each run's wall-clock budget in all four GRPO-labelled logs. Sparse and dense legacy positive-only variants suggest that dose changes whether gains are banked in mean success or multi-sample coverage, but that comparison is likewise exploratory.

Takeaway. The neural archive suggests estimator-associated coverage behavior and locates it in the easy/mid band, but it uses legacy allocation and recycling and its unbalanced adaptive schedules do not identify an interaction or validate the proposed loop.

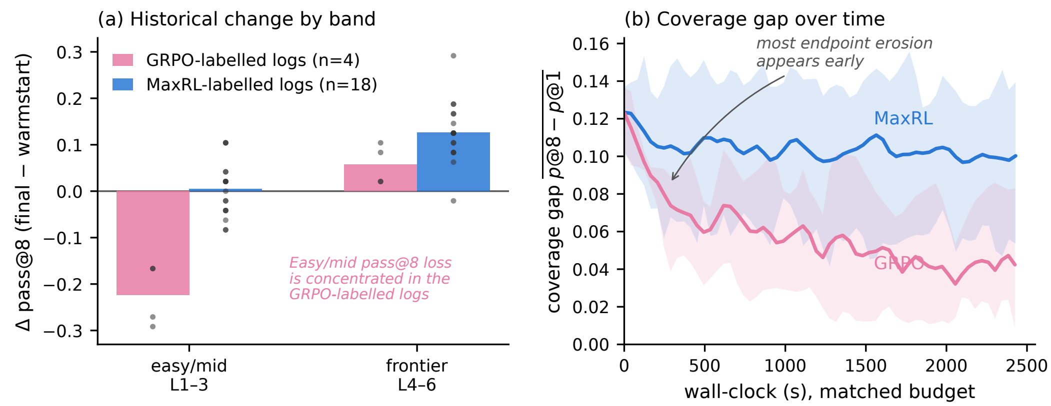

Figure 4: **Exploratory maze localization (matched wall-clock).** *(a)* Δpass@8 by difficulty band for a frozen archive of four GRPO and 18 heterogeneous practical-MaxRL runs. Dots are runs, not independent factorial replicates. *(b)* Mean and range of the coverage gap $\overline{\mathrm{pass@8} - \mathrm{pass@1}}$ over wall-clock. The figure localizes a historical association; it carries no pooled permutation test.

6.5 When does allocation *not* pay? (IsaacLab legged locomotion, one-seed pilot). A 5-arm Anymal-C rough-terrain pilot (reported by an independent team from a co-designed protocol; raw logs are not included here; App. A) is consistent with its anticipated null: on a terrain grid where every level is learnable at budget, the stock greedy curriculum beats the frontier teacher outright (mean pass .372 vs. .278). One seed cannot establish a regime rule; it supplies a boundary hypothesis for replication: when the pool is learnable throughout at the deployed budget, breadth may dominate frontier focus.

6.6 What does a curriculum add beyond task spread? (Gymnasium control — binary-success RLVR on classic control; official-task currency). All arms train on *binary task success* (MountainCar: reach the flag; CartPole: survive 400) — the classic-control analogue of a verifier reward. The **target-only** arm remains empirically frozen in the committed runs: no hardest-task success is observed, and any all-fail group has exactly zero weights under the deployed practical estimator. The artifacts do not record every parameter update, so we do not elevate that empirical freeze to a universal impossibility claim. The gated full stack under the identical reward reaches **1.000±0.000 on both official tasks** (3 seeds, Fig. 6). A dense-reward control confirms the zero is a property of sparse verifier-style reward (REINFORCE on CartPole's native reward solves it; App. A). Attribution at a common horizon: uniform-over-bins task spread supplies most of the effect (existence of a gradient). Recycling without allocation plateaus below ceiling on CartPole, while on MountainCar its final success reaches ceiling but later than the gated stack. Relative to uniform, the full gated stack is associated with speed and stability: roughly 3× faster to MountainCar hard-task 0.9 and, for common-280 hard-task AUC, about 6× smaller cross-seed sample SD than uniform on both tasks. A methods note from this control: our own first version used per-bin task-partitioned policy parameters and reproduced the known failure (0.000 in every arm). In this Gym setting, the curricula required shared parameters to transfer competence across bins.

Takeaway. In this control, task spread creates the first usable practical- estimator update, and shared parameters let that update transfer. Full CV is a separate all-fail baseline rather than part of this claim.

6.7 GSM8K 360M: a two-seed legacy-utility treatment-delivery pilot. The 50-step 2 × 2 study is much smaller than the MaxRL baseline's large-model evidence and is not confirmatory. In the registered seed, GRPO+teacher ends below GRPO uniform by −.027 mean@4 and −.048 pass@4; its second-half change (.096 → .093) is smaller than fixed-checkpoint evaluation noise. In seed 2 the endpoint differences have the same signs (−.007, −.026), but GRPO+teacher climbs throughout. Thus the registered regression shape occurs in one of two seeds. The .0094 repeated-evaluation SD

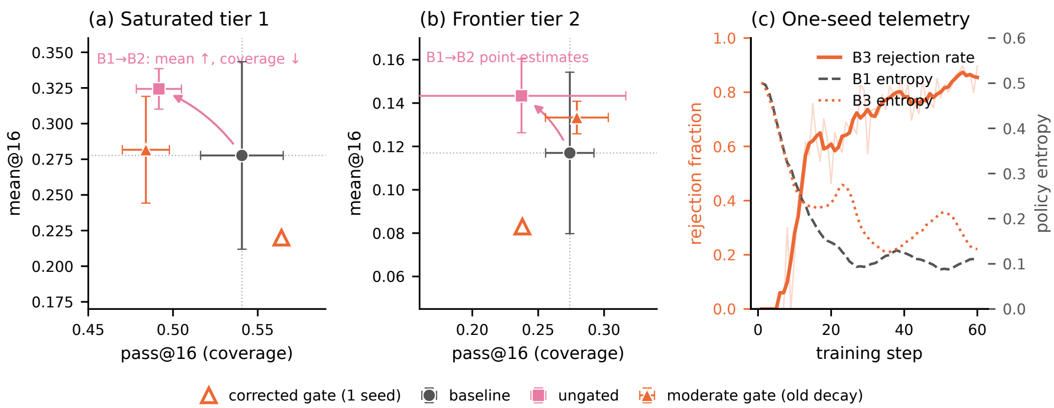

Figure 5: **Legacy per-trace Countdown sharpening and gate evidence.** *(a–b)* Three-seed endpoints for saturated tier 1 and frontier tier 2. Ungated legacy recycling (B2) raises mean success and lowers pass@16. The unintended moderate gate setting (B3) returns frontier-tier coverage but not saturated-tier coverage. The corrected stronger setting is shown as one seed, not as an error bar or a dose-response curve. *(c)* In-run rejection and entropy telemetry for one B3 seed.

estimates generation noise for one checkpoint; it is not training-seed uncertainty and is not used as a replication test.

The treatment itself uses the historical $(1-q^N)q$ utility rather than the derived ν_N profile and is weak and confounded. Teacher cells sample with replacement, so repetition differs from uniform; the replacement-matched control is incomplete. Across 3,200 groups and 7,473 prompts, only 2,561 prompts (34.3%) are visited: 0.428 group visits per pool prompt, or 1.25 visits per visited prompt. The posterior utility remains nearly flat. Teacher and uniform sampled-dead fractions have similar observed run means .66–.67; the more negative seed has a slightly stronger minimum separation. Under practical MaxRL, the single recorded teacher run finishes at .102 versus .108 for uniform, a null at this budget. We retain the pilot because it diagnoses posterior starvation and treatment delivery, not because it identifies a curriculum-by-estimator interaction. A replacement-matched, steering-controlled, multi-seed study with a frozen evaluation harness is required.

Takeaway. At this scale the legacy-utility prompt-level posterior is starved; the factorial interaction and the proposed sampler remain unresolved.

6.8 Countdown: a legacy per-trace recycler raises the mean while reducing coverage. The first pool was too easy: its baseline reached 76–94% of a random-guesser bound, so registered allocation and creation improvements returned null. It nevertheless exposed the relevant metric mismatch: relabeling improved mean@16 while reducing pass@16. We rebuilt the pool with operand permutation, parenthesization, exact division, and a headroom screen, then ran no recycling (B1), ungated recycling (B2), and a destination gate (B3) for three paired seeds. These historical runs assign each admitted trace its own achieved target within a uid group and use positive-only relabeled updates. They therefore have mixed conditioning and do not instantiate Algorithm 1's one-destination K'-of-N contrast. The results below describe that legacy recycler, not the proposed destination-group estimator.

The three-seed aggregate endpoint pattern recurs on the rebuilt pool. On tier 1, B1 to B2 changes mean@16 from $.278 \pm .066$ to $.324 \pm .014$ and pass@16 from $.541 \pm .024$ to $.492 \pm .013$. On frontier tier 2, mean changes from $.117 \pm .037$ to $.143 \pm .017$, while coverage changes from $.274 \pm .018$ to $.237 \pm .079$. These are aggregate mean and sample-SD summaries; the underlying three per-seed endpoint rows are not vendored, so we do not claim a per-seed direction. Only two of the three tier-2 pairs retain checkpoint trajectories. Both have higher recycled-arm mean at step 30, but their coverage paths differ: one is lower by step 15 and the other only by step 45. We therefore make no general timing or early-stopping claim.

This pattern is not a consequence of coefficient mass alone. The legacy achieved-destination selector samples destinations from the policy's own occupancy, so already-reachable outcomes are replayed more often; repeated selection is size-biased toward them relative to the task distribution. Exact verification preserves label validity but not destination balance. Related concentration mechanisms motivate GoalGAN, CHER, Skew-Fit, GCSL, and HIPI (Florensa et al., 2018; Fang et al., 2019; Pong et al., 2020; Ghosh et al., 2021; Eysenbach et al., 2020). We retain this as a direct measurement of a legacy RLVR recycler's large-k coverage cost. It does not establish the behavior of shared-anchor, contrastive relabeling.

6.9 Destination saturation is promising for the legacy recycler, but the gate is not yet a validated dial. At the unintended three-seed B3 operating point, tier-2 coverage is $.279 \pm .024$, close in point estimate to B1's $.274 \pm .018$, while mean@16 is $.133 \pm .008$, retaining roughly 60% of B2's mean-increment point estimate. On saturated tier 1, however, B3 is $.282 \pm .037$ mean / $.484 \pm .014$ coverage: it removes most of B2's mean change without restoring B1 coverage. A decay-math bug made these B3 runs weaker than the intended gate and turns them into an unintended moderate operating point.

The corrected stronger gate has only one seed: tier 1 $.220/.564$ (mean/coverage) and tier 2 $.083/.238$. It improves saturated-tier coverage while reducing both frontier metrics; together with B1–B3, this does not establish monotonicity or a Pareto frontier. One-run telemetry is mechanistically consistent with saturation admission—rejection rises from roughly $.12$ to $.85$ as destinations become reachable, and B3 retains more entropy than B1 in that seed—but it is not independent replication. The supported conclusion is narrower: destination pass rate is a plausible admission variable for this legacy relabeler, and one moderate setting returns the frontier-coverage point estimate near baseline. This is not a prespecified noninferiority result or evidence for the shared-anchor algorithm. A corrected fixed-decay multi-seed sweep with a prespecified coverage noninferiority test is required before recommending a gate strength.

Takeaway. Recycling-induced sharpening is a three-seed aggregate endpoint pattern for the legacy per-trace recycler. Its gate has a promising frontier operating point, but neither result validates the proposed common-destination algorithm.

6 RELATED WORK

Group estimators and difficulty weighting. GRPO originates in DeepSeekMath, while RLOO revisits leave-one-out REINFORCE-style optimization for language-model alignment (Shao et al., 2024; Ahmadian et al., 2024). MaxRL derives a family of likelihood objectives and unbiased finite-sample estimators (Tajwar et al., 2026); Mroueh independently analyzes GRPO's effective loss, calibration, and success amplification (Mroueh, 2025). SC-SDPO and A-GRAE make estimator–difficulty weighting explicit for GRPO-style training (Liu et al., 2026; Yu et al., 2026). Our distinction is finite-group and implementation-specific: raw, full-control-variate, and drop-constant-group MaxRL have different coefficient-mass profiles, and the last admits the exact factorization in Eq. 5. We then use that profile as a data-selection score and audit pass@k, rather than proposing another population objective.

Curricula and response-budget allocation. RLVR curricula include bucket bandits (DUMP), scalar difficulty controllers (AdaRFT), category bandits (SEC), dynamic filtering (DAPO, GRESO), and learnability-based prompt selection (SFL, LILO) (Wang et al., 2025; Shi et al., 2026; Chen et al., 2025; Yu et al., 2025; Zheng et al., 2025; Rutherford et al., 2024; Foster et al., 2025). LILO prioritizes success variance $p(1-p)$; Online Difficulty Filtering independently derives a policy-improvement lower bound favoring intermediate-difficulty tasks under GRPO (Bae et al., 2026). At $N = 2$, our normalized practical-MaxRL mass reduces to the same algebraic score; for $N > 2$, it is group-size- and estimator-specific. Reinforce-Ada reallocates a fixed batch-level response budget across prompts, including sequential allocation based on successes, while leaving macro prompt selection fixed (Xiong et al., 2025). Our teacher instead changes future prompt-group selection at fixed N. Proposition 3 is only the fixed-rate one-step allocation problem, not a general optimality claim.

Goal selection and hindsight proposal shift. HER is explicitly off-policy (Andrychowicz et al., 2017). GoalGAN generates intermediate-difficulty goals; HGG balances achievability against target relevance; CHER schedules relabel goals using proximity and diversity; and Skew-Fit uses inverse-density skew to improve state coverage (Florensa et al., 2018; Ren et al., 2019; Fang et al., 2019; Pong et al., 2020). GCSL derives an off-policy lower bound, while HIPI derives a corrected inverse-inference relabel law (Ghosh et al., 2021; Eysenbach et al., 2020). Difficulty- and coverage-aware relabel admission is therefore established. Our gate belongs to this family: the practical estimator's saturated endpoint motivates destination pass rate as the admission variable, but its threshold is tuned. The distinct empirical question is whether this admission preserves large-k coverage.

Hindsight in language and verifiable-reward loops. CodeIt and proof-tree relabeling reuse verified outputs for program synthesis and formal mathematics (Butt et al., 2024; Poesia et al., 2024). HiR directly rewrites failed instruction-following rollouts into hindsight instructions for RL; HSL and AgentHER relabel language-agent trajectories for supervised or preference training (Zhang et al., 2026; Li et al., 2026; Ding, 2026). LfH proposes hindsight instructions for VLA post-training, and ZipRL replays hindsight responses in multi-turn training (Xu et al., 2026; Hu et al., 2026). Thus language-agent relabeling is prior art. Related work also documents precision–coverage tension in RLVR and self-training (Yue et al., 2025; Cui et al., 2025; Yuan et al., 2026; Strozzi, 2026). Our focus is the conjunction: to our knowledge, no prior method (i) derives the exact finite-N coefficient-mass profile of deployed drop-constant-group MaxRL, (ii) uses that profile for cross-prompt allocation while using its saturation endpoint for destination admission, and (iii) evaluates the relabeling policy with large-k coverage. This conjunction is the proposed method; the current neural and LLM archives instantiate neither piece together and therefore do not provide end-to-end validation.

7 LIMITATIONS AND NEGATIVE RESULTS

Identification and scale. The maze archive reuses warm starts, mixes samplers, and uses a legacy utility/recycler, so it supplies a descriptive estimator association, not an interaction estimate or method validation. GSM8K is a 50-step 360M-parameter pilot with one or two seeds per cell, a replacement confound, and weak delivered steering. IsaacLab has one seed. The corrected strong gate has one seed. None supports a broad safety or scale claim; the required studies are an exact-ν_N, shared-anchor maze factorial; a replacement-matched multi-seed LLM factorial; independent locomotion seeds; and a corrected shared-anchor multi-seed Countdown gate sweep.

Estimator activity is not task transfer. ν_N exactly scales a prompt-local expected update, not its norm, SNR, cross-task benefit, or long-horizon value. The full-CV control is likewise local: under a common learning rate and matched budget, negative-only updates do not reliably unlock the exact-score chain. The post-hoc learning-rate sweep has one of 20 threshold crossings at two of ten rates and none elsewhere, so this result neither proves that full CV always fails nor supplies a reliable operating region. A separate fixed-schedule, held-out-tuned tabular control puts practical MaxRL, GRPO, and RLOO within .005 evaluation-grid AUC, but does not identify the neural curriculum interaction.

Hindsight is semantically verified but generally off-policy. The exact-score chain and corrected NumPy adapters instantiate the proposed shared-destination contrast; historical maze and Countdown runs do not. An exact verifier removes label error, not source-to-destination proposal shift. Same-group destination choice adds selection dependence, and our behavioral placebo battery does not estimate that bias. Cross-fitted destination selection, fresh-destination comparisons, and proposal diagnostics are open. Noisy or learned verifiers would add a second error source not studied here.

Pool support and shared transfer remain boundary hypotheses. The committed Jugs feasibility screen finds one shallow learnable stratum and zero observed success in deeper strata at its evaluation budget, despite nonzero relabel yield. The planned multi-arm training results are not vendored, so we do not infer a fixed point, entropy collapse, or a causal support requirement from that suite. No exact-score streaming control is vendored either, so the role of reusable cross-task transfer remains open.

Negative transfers and artifact limits. One longer-duration maze run did not solve the deep frontier. Project notes also describe single-seed γ-concentration, a $(1-p)$-tilted utility, and teacher posterior inflation from relabels, but their complete verdict artifacts are not vendored and we draw no quantitative conclusion from them. A historical adaptive-truncation result is withdrawn because its implementation dropped the $K = 0$ control variate; the corrected subset estimator has exact tests but no committed schedule-matched empirical comparison. Historical figures were assembled before a single frozen run manifest existed; a checked-in registry now freezes the Figure 4 cohort, while missing raw LLM and multi-seed corrected-gate artifacts prevent stronger reanalysis. The new estimator-control artifacts record seeds, commands, versions, source hashes, paired contrasts, and matched budgets, but the same standard must be applied prospectively to all neural runs.

8 Conclusion

The practical drop-constant-group MaxRL estimator has an exact finite-group activity profile, and its expected update factorizes into that scalar and a task-specific score separation. This result guides where allocation can seek contrast, while the three-estimator taxonomy prevents overextending it: full control variates can assign nonzero coefficients to all-fail groups, and hindsight changes the data proposal rather than merely the coefficient. The new 20-seed control shows that negative-only coefficients do not reliably bootstrap one extreme frontier at the matched budget, whereas exact shared-prefix relabeling does. Legacy Countdown then shows a possible cost: its per-trace positive-only recycler raises mean success while reducing multi-sample coverage. Because the neural and LLM runs predate both defining pieces of the proposed loop, end-to-end method validation remains open. The practical recommendation is therefore modest: state the exact estimator implementation, audit finite-group activity, separate local activity from transfer, and report pass@k alongside the mean.

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A Full details for compressed experiment rungs

Material moved from §6 during the page-budget pass. The new estimator controls and plotted aggregate endpoints are committed with the paper; historical evidence with incomplete run-level artifacts is identified below.

6.1 control battery (full). Eight arms, 5 seeds each: uniform 0.650, lr$\times 2$ 0.798, random-target relabels 0.800, live-gradient replay 0.832, true relabels 0.869, no-floor γ-matched oracle $0.8885\pm.0015$, full stack $0.8895\pm.0026$, oracle+recycling $0.8935\pm.0013$. The relabel hindsight-minus-replay difference $(+.037)$ has paired seed spread $\pm.0136$; its worst seed beats every placebo's best. Random-target relabels are lower than live-gradient replay $(-.032$, 5/5 seeds) but neutral vs. plain lr-doubling. This arm reused a stale deepest-prefix reward vector, drew destinations from an unseeded RNG, and did not exactly re-verify them; per-arm update counts and norms were not logged. It is therefore a historical behavioral placebo, not a clean dose-matched or direction-isolating control. Utility-form controls: exact ν_N $0.728\pm.002$ vs. legacy frontier form $0.733\pm.015$ (the derived form's payoff has about $7\times$ smaller sample SD); the $N{=}2$ slice $p(1{-}p)$ is a wash against uniform $(-.007$, 2/5 wins) where ν_N carries $+.078$ (5/5). AUC spread across arms is $9\times$ the final-performance spread.

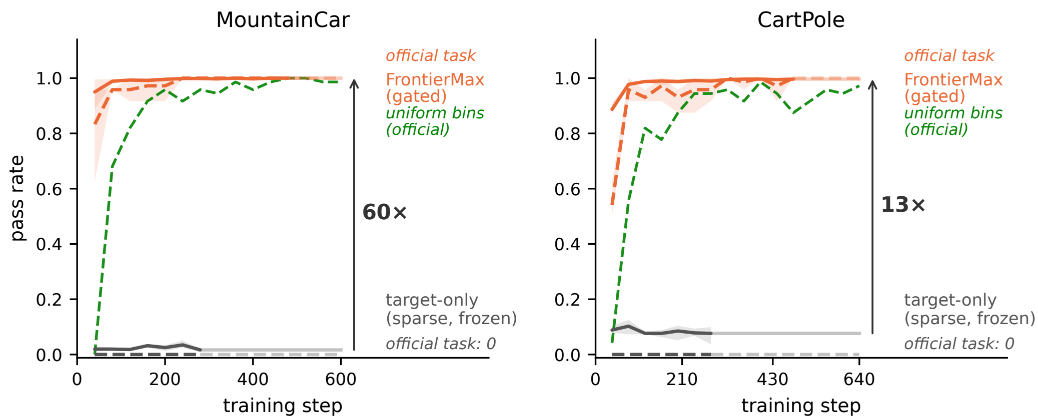

Figure 6: **Gym control under binary task-success reward, trained to plateau (3 seeds).** Solid: mean pass rate across curriculum bins; dashed: the *official* task (MountainCar flag / CartPole survive-400). Under the practical drop-constant-group estimator, the target-only arm (gray) remains empirically frozen; all-fail groups receive zero weights. A uniform-over-bins control (green) cracks both tasks but does not end at 1.000 in every seed; on MountainCar it is roughly 3× slower to hard-task 0.9. The gated FrontierMax stack converges to 1.000 ± 0.000.

6.5 IsaacLab protocol. According to the independent team's one-seed fixed-grid pilot summary (raw logs are not included here), the five arms are: frontier teacher, stock greedy walker, uniform, scripted ramp, no-motion control; fixed-grid per-level evaluation; Anymal-C rough terrain; one seed, executed by an independent team against a co-designed protocol. The teacher's only edge over the greedy walker is a small reported difference at the hardest evaluated level; no raw logs are available to estimate its noise.

6.6 gym control (full). A dense-reward control using CartPole's native $+1$/step reward lets vanilla REINFORCE on the same tile policy solve survive-400 (0.963 ± 0.038, 3 seeds); MountainCar's native -1/step reward (constant until first success) stays at 0.000 in all seeds — the categorical zero tracks reward sparsity, not the environment. Attribution at a common 280-step horizon (hard-task AUC): MC .750 → +recycling .895 → +teacher .956; CartPole .708 → .792 → .893; paired teacher increment positive 3/3 seeds on both; creation-without-allocation plateaus at 0.958 on CartPole in every seed while adding the teacher restores 1.000; teacher speed effect $3.0\times$ to hard-task 0.9 ($2.1\times$ to 0.99 on CartPole). Earlier random-policy probes and a transition-matched study are documented in project notes but are omitted here because their raw artifacts are not vendored.

6.3 inference-efficiency detail. Convention: smallest k reaching absolute coverage 0.25, log-interpolated. Artifact `efficiency.json` is under `curriculum_maxrl/maze_gpu/`; Fig. 7 plots the summed-coverage crossing. Summed coverage over L0–6: GRPO 3.24 vs. MaxRL 3.07 at pass@1; crossing at $k{\approx}4$. At $k{=}64$, MaxRL reaches 5.125 versus GRPO's 4.625; GRPO's level-2/4 curves reach $0.875/0.5625$, versus $1.000/0.6875$ for MaxRL.

6.9 secondary read (exploratory). The committed training trajectories contain complete entropy and gate-rejection telemetry for one B1/B2/B3 seed. Over its final 15 steps, mean entropy is $.099/.067/.181$, respectively. B3's rejection rate trends from roughly .12 early to .86 late, although the path is not monotone. These one-run telemetry summaries are mechanistic diagnostics, not independent replication.

6.7 artifact and harness caveats. The committed cell trajectories and pass@k summary reproduce the reported training curves and endpoint sweep. The sweep's absolute level is roughly three times below the trainer validation metric; that harness discrepancy is unresolved, so only within-sweep contrasts are used. Posterior-agreement, external-difficulty, response-length, and prompt-uniqueness

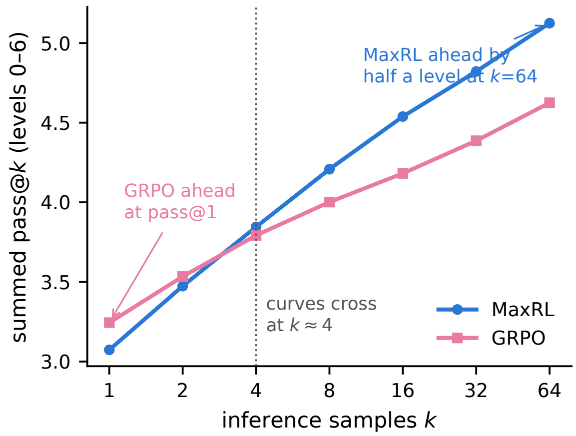

Figure 7: **Summed maze coverage vs. inference samples k (final uniform-arm checkpoints).** In this checkpoint pair, GRPO leads at pass@1, the curves cross at $k{\approx}4$, and practical MaxRL leads by half a level at $k{=}64$. This is a secondary diagnostic, not a general estimator-selection rule.

diagnostics in the historical postmortem require non-vendored checkpoints or annotation caches and are therefore not used as paper evidence.

Jugs feasibility screen (§7). The committed screen evaluates the first 40 tasks per tier from a committed 200-task-per-tier pool t0–t4 (2–5 jugs, disjoint minimum-move bands), using SmolLM2-360M two-shot generation and $N = 16$. It records pass@1/pass@16 of $.058/.475$ on t0, $.006/.100$ on t1, and $0/0$ on t2–t4, while failure-to-relabel yield is .62–.73 across tiers. No B1/B2/B3 training-result artifact is present in this checkout. Fixed-point, entropy, relabel-rate, and gate-rejection claims are therefore omitted until a run-level snapshot is vendored.

B Hyperparameters and Compute

Project notes estimate approximately 230 A10G-hours for the historical GPU studies, plus CPU time for the tabular suites; incomplete per-session accounting prevents reconstructing that total from committed artifacts. Across 1,188 mixed GSM8K and Countdown telemetry steps, the step-weighted mean generation fraction is 31.2%. Suite-specific raw timing logs are not vendored, so this aggregate is not a representative per-suite cost breakdown. Table 3 gives the per-suite configuration and Table 4 the knob inventory with provenance.

Table 3: Per-suite configuration. These are the principal settings for each suite; tuning and evidence status are separated in Table 4.

	chains (§6.1–6.2)	maze (§6.3–6.4)	GSM8K (§6.7)	Countdown (§6.8–6.9)
policy	exact-score tabular	1.26M transformer	SmolLM2-360M	SmolLM2-360M (SFT)
allocation/recycler	ν_N / shared anchor	legacy frontier-ALP / isolated positives	legacy utility / none	legacy utility / per-trace targets
rollouts N	16	32	16	16
groups/step	1 new; 8 legacy	8	64	64
optimizer, lr	SGD; .5 common; tuned 12/16/64 (MaxRL/GRPO/RLOO)	AdamW, 10^{-4}	AdamW, 10^{-5}	AdamW, 10^{-5}
budget	3,200 groups	2,400 s clock	50 steps	60 steps
seeds (headline)	20 primary; 10 tune + 20 held-out; 5 legacy	3 selected/arm	1–2/cell	3/arm; 1 corrected
compute/run	CPU-min	$\sim$0.7 A10G-h	$\sim$20 A10G-h	$\sim$6 A10G-h

Table 4: Teacher and recycler knobs: value, provenance, and measured sensitivity.

knob	default	derived or tuned	sensitivity
profile shape/peak/N-dependence	—	**derived** (Prop. 1)	the theory claim
posterior decay	0.7	tuned (V2b)	historical sensitivity; raw artifact absent
uniform floor	0.1	tuned	historical sensitivity; raw artifact absent
γ (concentration)	1	tuned (V6)	4 on chains; single-seed maze note, complete verdict artifact absent
`gate_max_p`	0.5	tuned; saturation-motivated	corrected sweep remains open
relabel dose cap	12 Countdown groups; 16 maze trajectories	suite-specific tuned values	no cap in corrected core; isolated dose response remains open

Implementation contract and reproducibility

The finite-group derivations assume i.i.d. rollouts conditional on a requested task and a binary verifier. Centering statistics are computed over the full requested group; GRPO uses the sample standard deviation plus the stated stabilizer. Hindsight selects one common destination for an all-fail group, rewrites every trajectory to that destination, re-verifies the rewritten group, and updates only when $0 < K' < N$ in the proposed contract. Its teacher records evidence for the requested task, not the relabeled destination. GridReach and corrected Countdown regression-test shared-destination contrast; chain and Gym exercise relabeling without dedicated assertions for all three contracts. GridReach and Countdown expose only bucket-level posterior IDs, so their optional gates are bucket rather than exact-destination estimates. Historical maze and Countdown runs use the legacy semantics stated in Table 3 and are not tests of this contract.

The new finite-group control uses `curriculum_maxrl/run_estimator_variants.py` with 20 seeds; the post-hoc robustness sweep uses `curriculum_maxrl/run_fullcv_lr_sensitivity.py`; and the held-out fixed-schedule comparison uses `curriculum_maxrl/run_schedule_matched_estimators.py`. Structured child random streams keep rollout and schedule randomness separate across replicates while preserving common random numbers within paired arms. Their JSON outputs record seed identities, matched budgets, versions, commands, source hashes, and paired contrasts. Analytic curves and the new control panels regenerate from committed code and JSON. Figure 4 loads its 22-row frozen registry; other historical neural endpoints are plotted from available aggregate or run logs, but no single manifest predates all of those runs. Raw GSM8K, multi-seed corrected-gate, IsaacLab, and complete Jugs run-level archives remain gaps. The dependency-light `frontier_rl` NumPy core and exact-score testbed are runnable here; its LLM, IsaacLab, and VLA integrations are prototypes and are not all end-to-end reproducible in this checkout.